

IIT Madras BS Data Science (Foundation): Deep Analysis of Mathematics for Data Science 2 (BSMA1003) & Statistics for Data Science 2 (BSMA1004)

TL;DR

- **Maths 2 (BSMA1003)** is an 11-week linear-algebra-and-multivariable-calculus course (vectors, matrices, vector spaces, linear transformations, inner products, Gram-Schmidt, partial derivatives, Hessian/optimization); **Stats 2 (BSMA1004)** is a 12-week inferential-statistics course (joint/multiple random variables, continuous distributions, estimation/MLE/method of moments, Bayesian estimation, confidence intervals, hypothesis testing, regression). Both are 4-credit Foundation courses.
- The biggest structural change to know: **from the January 2026 term, GAA (weekly assignment) weightage in all Foundation courses was set to 0**; instead 10 marks inside the two quizzes and end-term are drawn from the assignments. Both courses now use the formula $T = \max(0.6F + 0.3 \cdot \max(Qz1, Qz2), 0.45F + 0.25 \cdot Qz1 + 0.3 \cdot Qz2)$ plus a small course bonus, so the final exam dominates the grade.
- Both courses are widely regarded as the two hardest Foundation courses; Maths 2's abstract vector-space/linear-transformation material and Stats 2's estimation/hypothesis-testing density are the main difficulty spikes. Watch the official course pages on study.iitm.ac.in and Prof. Sarang Sane's (Maths 2) and Prof. Andrew Thangaraj's (Stats 2) YouTube playlists as the primary resources.

Key Findings

Course identity and placement. Both are 4-credit Foundational-level courses in the IITM BS in Data Science and Applications. Maths 2 (course code BSMA1003) has a prerequisite of Maths 1 (BSMA1001). Stats 2 (BSMA1004) has prerequisites of Stats 1 (BSMA1002) and Maths 1 (BSMA1001), and lists Maths 2 (BSMA1003) as a co-requisite — meaning Stats 2 is intended to be taken alongside or after Maths 2 because it leans on linear algebra and calculus. The Foundation level has 8 courses / 32 credits total; all 8 must be cleared before entering the Diploma level. [iitm + 3](#)

Run length. Maths 2 runs **11 weeks** of coursework; Stats 2 runs **12 weeks**. Both have weekly online assignments, 2 in-person invigilated quizzes, and 1 in-person invigilated end-term exam. [iitm](#)

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Instructors. Maths 2 is taught by **Prof. Sarang S. Sane** (Assistant Professor, Department of Mathematics, IIT Madras). Stats 2 is led by **Prof. Andrew Thangaraj** (Professor, Electrical Engineering Department, IIT Madras), supported by a team of instructors: **Prashant Sharma, Prerna Gupta, and Swathy S. Sajeev** (all M.Sc. Statistics, University of Delhi).

Details

A. Mathematics for Data Science 2 (BSMA1003) — Official Week-by-Week Syllabus

The course's stated aim is to introduce linear algebra, calculus and optimization with a focus on machine learning and data science applications. The official 11-week breakdown: [iitm](#)

- **Week 1 — Vectors and matrices:** Vectors; Matrices; Systems of linear equations; Determinants (parts 1 & 2). [iitm](#)
- **Week 2 — Solving linear equations:** Determinants (part 3); Cramer's Rule; solutions to a system of linear equations with an invertible coefficient matrix; the echelon form; row reduction; the Gaussian elimination method. [iitm](#)
- **Week 3 — Introduction to vector spaces:** Introduction to vector spaces; properties of vector spaces; linear dependence; linear independence (parts 1 & 2). [iitm](#)
- **Week 4 — Basis and dimension:** What is a basis for a vector space; finding bases for vector spaces; rank/dimension of a vector space; rank and dimension using Gaussian elimination. [iitm](#)
- **Week 5 — Rank and nullity; intro to linear transformation:** The null space of a matrix — finding nullity and a basis (parts 1 & 2); what is a linear mapping (parts 1 & 2); what is a linear transformation. [iitm](#)
- **Week 6 — Linear transformation, kernel and images:** Linear transformations, ordered bases and matrices; image and kernel of linear transformations; examples of finding bases for the kernel and image. [iitm](#)
- **Week 7 — Equivalent and similar matrices; intro to inner products:** Equivalence and similarity of matrices; affine subspaces and affine mappings; lengths and angles; inner products and norms on a vector space. [iitm](#)
- **Week 8 — Orthogonality, orthonormality, Gram-Schmidt:** Orthogonality and linear independence; orthonormal basis; projections using inner products; the Gram-Schmidt process; orthogonal transformations and rotations. [iitm](#)
- **Week 9 — Multivariable functions, partial derivatives, limits/continuity/directional derivatives:** Multivariable function visualization; partial derivatives; directional derivatives; limits and continuity for scalar-valued multivariable functions; directional derivatives in terms of the gradient. [iitm](#)
- **Week 10 — Directional ascent/descent, tangent (hyper)plane, critical points:** Direction of steepest ascent/descent; tangents for scalar-valued multivariable functions; finding the tangent hyperplane; critical points for multivariable functions. [iitm](#)
- **Week 11 — Higher-order partial derivatives, Hessian and local extrema, differentiability:** Higher-order partial derivatives and the Hessian matrix; Hessian and local extrema for $f(x,y)$ and $f(x,y,z)$; differentiability for multivariable functions; review of Maths 2.

Stated learning outcomes: manipulating matrices with matrix algebra; elementary row operations; Gaussian elimination to solve linear systems; testing linear independence and writing dependencies; finding subspaces with bases and ranks; distances and angles via norms and inner products; orthonormal bases via Gram-Schmidt; finding maxima/minima of single-variable functions using derivatives and of multivariate functions using vector calculus. (IIT Madras)

Note on eigenvalues: Although eigenvalues/eigenvectors are core to linear algebra and appear in some reference materials, they are **not** an explicit week on the current official BSMA1003 page — the course emphasizes vector spaces, linear transformations, inner products and multivariable calculus/optimization instead. Eigenvalue material is taken up more in later (Diploma/Degree, e.g. Machine Learning) courses.

B. Statistics for Data Science 2 (BSMA1004) — Official Week-by-Week Syllabus

This course builds on Stats 1 and develops the main statistical inference problems and solution approaches. The official 12-week breakdown:

- **Week 1 — Multiple random variables:** Two random variables; multiple random variables and distributions (joint distributions). (iitm)
- **Week 2 — Multiple random variables (cont.):** Independence; functions of random variables — visualization; functions of multiple random variables. (iitm)
- **Week 3 — Expectations:** Casino math; expected value of a random variable; scatter plots and spread; variance and standard deviation; covariance and correlation; inequalities. (iitm)
- **Week 4 — Continuous random variables:** Discrete vs. continuous; weight data; density functions; expectations. (iitm)
- **Week 5 — Multiple continuous random variables & probability models:** Height/weight data; two continuous random variables; averages of random variables (Colab illustration); limit theorems; IPL data — histograms and approximate distributions; jointly Gaussian random variables; probability models for data (simple models, models based on other distributions, models with multiple random variables and dependency, models for IPL powerplay, models from data). (iitm)
- **Week 6 — Refresher week.** (iitm)
- **Week 7 — Estimation and Inference I:** Statistical inference and the estimation problem; parameters vs. statistics; random/i.i.d. sampling; sampling distributions of statistics (sample mean, sample variance); point estimation; properties of estimators — unbiasedness, bias, consistency, efficiency, mean squared error; method of moments estimation. (Google)
- **Week 8 — Estimation and Inference II:** Likelihood function and maximum likelihood estimation (MLE); log-likelihood; MLE for standard distributions; interval estimation / confidence intervals for means and proportions; the bootstrap method for approximate confidence intervals. (Google)
- **Week 9 — Bayesian estimation:** Prior distribution, likelihood, posterior distribution ($\text{posterior} \propto \text{prior} \times \text{likelihood}$); conjugate priors; Bayesian point estimates (posterior mean /

MAP); Bayesian vs. frequentist contrast.

- **Week 10 — Hypothesis testing I:** Null and alternative hypotheses (H_0 , H_1); simple vs. composite hypotheses; test statistic, rejection region, level of significance (α); Type I and Type II errors (β); power of a test ($1-\beta$); Z-test; p-values and decision rules. (Studocu)
- **Week 11 — Hypothesis Testing II:** t-test (one-sample, variance unknown); two-sample tests (difference of means); tests for variance using chi-square (χ^2) and tests for proportions; connection between confidence intervals and hypothesis tests; setting up hypothesis tests for simple linear regression.
- **Week 12 — Revision week.** (iitm)

Stated learning outcomes: recalling statistical modeling and description of data; applying probability distributions to datasets; explaining estimation of parameters; solving point and interval estimation problems; testing of hypotheses related to mean and variance; analysing data using simple regression models and setting up relevant hypothesis tests. (IIT Madras)

Sourcing caveat: The official page lists only the vague titles for Weeks 7–12; the detailed subtopics above are assembled from the official learning outcomes, IITM student-portfolio course pages, and IITM-specific student notes (Studocu). The exact week placement of a given concept (e.g. MLE vs. Bayesian) can shift by ± 1 week between term runs.

C. Course Structure, Credits and Curriculum Fit

Each Foundation term runs ~12 weeks of coursework plus quizzes and end-term. A learner registers for 1–4 courses per term depending on their Credit Clearing Capability (CCC), which in the Foundation level is based on the Qualifier Exam or the previous term's end-term results. The Foundation level (8 courses / 32 credits) must be fully cleared before entering the Diploma level. Stats 2 and Maths 2 feed directly into Diploma Data Science courses — particularly Machine Learning Foundations, Machine Learning Techniques and Business Analytics — which assume comfort with linear algebra, probability distributions, estimation and hypothesis testing. (IIT Madras) (IIT Madras)

D. Assessment / Exam Structure and Grading Formula

Standard Foundation assessment components: weekly graded online assignments (GAA), Quiz 1, Quiz 2 (both in-person invigilated), and an in-person invigilated end-term exam (1.5 hours).

Eligibility to write the end-term exam (both courses): appear for at least one of the two quizzes **AND** meet the assignment-average threshold of $\geq 40/100$. *Note a sourcing discrepancy here:* the January 2026 course-specific grading document states the threshold as the **average of the best 5 out of the first 7 weeks** of weekly assignment scores, whereas the general Academics page states the **best 5 out of the first 9** weekly assignments. Both require $\geq 40/100$ plus one quiz attended; treat the stricter/earlier "first 7 weeks" interpretation as the safe planning assumption and confirm against your current term's grading document. If eligibility is not met, the learner cannot sit the end-term and must repeat the entire course. **Eligibility to obtain the final course grade:** attending the end-term exam.

MAJOR CHANGE (effective January 2026 term): Weightage for GAA in all Foundation-level courses was made **0**. Instead, questions for a total of **10 marks** in the two quizzes and the end-term exam are drawn directly from the assignments. (For Diploma courses GAA weight became 5 marks; Degree-level unchanged.) [Google](#)

Current (Jan 2026) grading formula — Mathematics for Data Science 2 (BSMA1003):

$$T = \max(0.6F + 0.3 \cdot \max(Qz1, Qz2), 0.45F + 0.25 \cdot Qz1 + 0.3 \cdot Qz2) + B - \text{capped to } 100$$

[google](#)

where Qz1, Qz2 = quiz scores (0 if not attempted), F = final exam score, and B = bonus (6 marks total, from 3 course-specific activity assignments of 2 marks each, added only if the student passes). [google](#)

Current (Jan 2026) grading formula — Statistics for Data Science 2 (BSMA1004):

$$T = \max(0.6F + 0.3 \cdot \max(Qz1, Qz2), 0.45F + 0.25 \cdot Qz1 + 0.3 \cdot Qz2) + \text{Extra-activity bonus (5 marks, capped)} - \text{capped to } 100$$

[google](#)

where the Stats 2 bonus is 5 marks total (3.75 marks for weekly extra activities + 1.25 marks based on activity quality at instructors' discretion), added only on passing. Stats 2 features 5 peer-reviewed Extra Activities. (These formulas are from the Jan 2026 term grading document; community grading tools such as the IITM BS Grade Suite likewise note they are "based on the Jan 2026 Term grading document ... Formulas may change each term.") [google](#)

Interpretation: Under the current scheme the **final exam (F) carries 45–60% of the grade depending on which branch of the max() is larger**, and the two quizzes carry the rest. The formula is student-friendly: it automatically takes whichever combination gives the higher score, so a strong final exam can compensate for one weak quiz. Because GAA weight is now 0, weekly assignments no longer directly add marks — but they remain critical because (a) they gate end-term eligibility and (b) 10 marks across quizzes/end-term are lifted directly from them.

Topic-to-assessment mapping (typical): Quiz 1 usually covers the first ~4 weeks; Quiz 2 covers roughly the middle weeks up to ~week 7–8; the end-term is cumulative across all weeks (Weeks 1–11 for Maths 2; 1–12 for Stats 2, with Week 11/12 content explicitly examinable even though those assignments don't count toward GAA). For Maths 2 this means Quiz 1 ≈ matrices/determinants/linear systems/vector spaces; Quiz 2 ≈ basis/rank/linear transformations/inner products; end-term ≈ everything including Gram-Schmidt and multivariable calculus/optimization. For Stats 2, Quiz 1 ≈ joint/multiple random variables, expectations, continuous distributions; Quiz 2 ≈ probability models and estimation/inference; end-term ≈ everything including Bayesian estimation and hypothesis testing.

Exam dates (Jan 2026 term, illustrative): Quiz 1 — March 15 2026; Quiz 2 — April 12 2026; End-term — May 10 2026. [google](#) [google](#)

Pass grading: The program uses absolute grading with cutoffs (S/A/B/C/D/E pass; U/WA/WQ fail; I = incomplete). Standard cutoffs are at 90/80/70/60/50/40. [google](#)

E. Textbooks, Reference Materials and Official Portals

Maths 2 (BSMA1003):

- Primary study material: course videos and assignments on the course portal.
- Official reference document: a downloadable "**Linear Algebra**" PDF linked on the course page. A widely used companion is "**A Short Course on Linear Algebra**" by M. Thamban Nair (IIT Madras). [IIT Madras](#)
- Official course page: study.iitm.ac.in/ds/course_pages/BSMA1003.html
- YouTube playlist: "Mathematics for Data Science II" (Prof. Sarang Sane). [YouTube](#)

Stats 2 (BSMA1004):

- Primary study material: course videos and assignments on the course portal.
- Official reference documents: downloadable "**Joint Discrete Distributions (Vol 1)**" and "**Joint Continuous Distributions (Vol 2)**" PDFs (linked on the official course page).
- Prescribed textbook: "**Probability and Statistics with Examples using R**" by Siva Athreya, Deepayan Sarkar and Steve Tanner. A secondary recommended text is "**Probability & Statistics for Engineers & Scientists (Global Edition)** by Ronald E. Walpole. [iitm](#)
- Official course page: study.iitm.ac.in/ds/course_pages/BSMA1004.html
- YouTube playlist: "Statistics for Data Science 2" (Prof. Andrew Thangaraj).

F. Practical Study Strategy, Difficulty and Resources

Difficulty. In student accounts, Maths 2 and Stats 2 are described as the two hardest Foundation courses. The letslearn1110.com student blog "My Experience of the Foundation Level in IIT Madras BS Degree" ranks **Mathematics 2 as the hardest course in the Foundation level** and states verbatim: "*Statistics 2: This was the hardest subject in the foundation level after Mathematics 2. It covered an extensive range of concepts, making it challenging to master.*" For Maths 2 the same student wrote: "*For challenging subjects like Mathematics 2, I focused on practicing daily and revisiting lecture recordings to clarify doubts.*"

Maths 2 difficulty hotspots: the conceptual jump from concrete matrix manipulation (Weeks 1–2) to abstract vector spaces, linear independence, basis/dimension and linear transformations (Weeks 3–6) is where students struggle most. Inner products, projections and Gram-Schmidt (Weeks 7–8) and the shift to multivariable calculus/Hessian-based optimization (Weeks 9–11) are also heavy. Strategy: master row reduction and rank early (it recurs throughout), build geometric intuition for vector spaces, and practice proof-style reasoning rather than rote computation since exam questions test understanding.

Stats 2 difficulty hotspots: joint/multiple random variables and jointly Gaussian variables (Weeks 1–5), and the estimation–inference–hypothesis-testing block (Weeks 7–11) where MLE, method of moments, Bayesian estimation and the family of tests (Z , t , χ^2) are introduced in rapid succession. Strategy: solidify the Stats 1 probability foundation first; keep a formula/results sheet for distributions, estimators and test statistics; practice with the R-based examples and the official Colab notebooks; and drill previous-year quiz/end-term questions.

Best resources students actually use:

- Official lecture videos (primary) and the weekly graded assignments (must-do for eligibility and because exam questions are drawn from them).
- Previous-year question papers (PYQs) for both courses — heavily used for end-term prep; several YouTube channels publish "Maths-2 End Term — important PYQs" walkthroughs.
- Community study-material repositories: Studocu (IITM-specific week-wise GA solutions and notes), theopennotes.in, iitmdatascience.com (notes & question papers), and student-built Google Sites portfolios under ds.study.iitm.ac.in.
- The IITM Discourse forum and the r/IITM_BS subreddit for doubt-solving and term-specific advice.
- Grade/CGPA calculators built by students (e.g. IITM Scholar Hub, MTR Calc, IITM BS Grade Suite) to model how quiz/final combinations affect the final T score.

Recommendations

Stage 1 — Before the term (both courses): Confirm prerequisites are solid. For Maths 2, revise Maths 1 functions/derivatives and basic matrix arithmetic. For Stats 2, revise Stats 1 probability, random variables, and Binomial/Normal distributions — and ideally take Maths 2 concurrently or first, since Stats 2 uses linear algebra and calculus.

Stage 2 — Weeks 1–7 (eligibility phase): Treat the weekly assignments as non-negotiable: you need the best-5 assignment average $\geq 40/100$ (over the first 7 weeks per the course-specific doc; confirm whether your term uses 7 or 9) plus one quiz attended to be allowed to sit the end-term. Do every assignment even though GAA weight is now 0, because 10 marks in the quizzes/end-term come straight from them. For Maths 2, do not let the Week 3–6 abstraction slide; for Stats 2, lock down joint distributions and expectations before estimation begins.

Stage 3 — Quiz preparation: Since the formula rewards $\max(Qz1, Qz2)$ heavily, aim to peak on at least one quiz. Use PYQs for the relevant week ranges. A weak quiz is recoverable, but attending at least one quiz is mandatory for eligibility.

Stage 4 — End-term (highest leverage): Because F carries 45–60% of the grade, the end-term is where the course is won or lost. Prepare all weeks including Week 11/12 content (explicitly examinable). Practice full-length PYQ papers under timed conditions. For Maths 2 prioritize Gaussian elimination/rank, linear transformations, Gram-Schmidt and Hessian-based optimization; for Stats 2

prioritize MLE/method of moments, confidence intervals, and the Z/t/ χ^2 test family with correct table usage.

Benchmarks that should change your strategy: If your best-5 assignment average is trending below ~50/100 by Week 5, escalate immediately (office hours, Discourse, tutoring) — you risk losing end-term eligibility. If you score below ~40 on Quiz 1, treat Quiz 2 and the final as recovery opportunities and lean on the max() branches. If you are time-constrained and must drop effort somewhere, protect end-term preparation over chasing marginal assignment perfection.

Caveats

- **Syllabus version sensitivity:** The week-by-week breakdowns above reflect the current official course pages (verified mid-2026). IITM periodically revises content; the official study.iitm.ac.in course pages are the authoritative source. Detailed Stats 2 Week 7–12 subtopics are reconstructed from official outcomes plus IITM student notes, and exact week placement of a concept can shift ± 1 week between runs.
- **Grading formula is term-specific:** The T-formulas and the GAA-weight-zero rule are from the January 2026 term grading document. Always check the current term's official grading document, since formulas, bonus structures and quiz counts change term to term.
- **End-term eligibility discrepancy:** Sources disagree on whether the assignment-average eligibility window is "best 5 of first 7" (Jan 2026 course-specific grading doc) or "best 5 of first 9" (general Academics page). Verify against your term's grading document.
- **Quiz/end-term topic mapping** is based on typical past patterns, not an official guarantee; exact coverage is announced each term.
- Some figures (difficulty rankings, "hardest course") come from individual student blogs/forums and reflect opinion, not official data.